

Corneal Infection

Guidelines for Clinical Case Presentations

Anoop Thomas, Mythri HM, Mallipatna C Ashwin, Shamin Jacob

HISTORY:

Presenting Complaints:

- Pain
- Redness
- Lacrimation

Details:

Pain

- Onset, Distribution, Character, Aggravating and Relieving Factors

Redness

- Associated Photophobia

Lacrimation

Associated History

- Coloured Haloes
- Dimness of Vision
- Lid Edema and Blepharospasm
- Foreign Body Sensation

Relevant History:

- Trauma
- Contact Lens Wear
- Cataract or Intraocular Surgery
- Topical Medication – specifically Steroids and Local Anaesthetic
- Alcoholism, Malnutrition, Immunosuppression
- Herpetic Skin Lesions
- Diabetes, Hypertension
- Similar Episodes in the Past
- Oral Medications

Family History

EXAMINATION:

General Physical Examination:

- Anaemia
- Nutritional Deficiency

Ocular Examination:

- Eyelid:
 - Lid closure, Apposition, Punctum
 - Margins, Trichiasis, Ectropion, Entropion
 - Meibomitis, Blepharitis
 - Marginal Tear Meniscus
 - ROPLAS
- Conjunctiva
 - Congestion
 - Limbal papillae
 - Upper Tarsal Conjunctiva:
(Foreign Body, Concretions, Follicles, Papillae, Scarring, Ulceration)

- Cornea
 - Size, Shape
 - Corneal Lustre
 - Transparency
 - Opacity
 - Corneal Edema
 - Vascularisation, Pannus
 - Corneal Nerves
 - Epithelial Defect
 - SPK, Filaments, Abrasions
 - Pigments / KP's
 - Corneal Sensation
- Look for Contact Lens
- Anterior Chamber
- Iris: synechiae
- Pupil: direct and consensual
- Lens
- Fundus

Systematic slit lamp examination of the cornea:

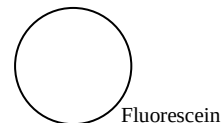
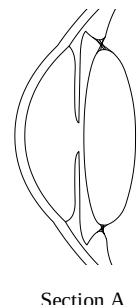
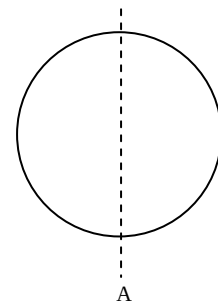
- *Diffuse*:
General survey and location of abnormality
- *Oscillating*:
General survey and location of abnormality
- *Sclerotic scatter*:
Pattern of abnormality
- *Optical section*:
Depth of opacity
- *Parallelepiped*:
Surface details
- *Proximal illumination*:
Internal details
- *Retroillumination*:
Optical quality of the cornea
- *Specular reflection*:
Surface irregularities and endothelial mosaic
- *Colour filters*:
Green – enhance vessels and pigments
- *Staining*:
Fluorescein (cobalt blue filter) and Rose Bengal

Description of ulcer:

- Site
- Size
- Shape
- Surface
- Edge
- Margin
- Surrounding tissue

CORNEAL DIAGRAM:

1. Basic outline
 - a. Draw circle (30mm diameter)
 - b. Add pupil (as a landmark) (in brown)
 - c. Indicate location of cross-section (dotted line)
 - d. Draw free hand of cross-section
2. Pattern of Pathology
 - a. Outline of opacity (blue) – use sclerotic scatter and broad tangential illumination
 - b. Draw a line to indicate the epithelium in the cross-section
3. Detail of Pathology
 - a. Refine the findings (scar, degeneration, infiltrate, oedema)
 - b. Add labels
 - c. Draw the findings in the appropriate levels in the cross-section.
4. Other corneal findings (vessels, pigments, posterior deposits)
5. AC and Lens in the cross-section
6. Vital staining (use a separate sketch if required)
7. Measurements to be written



COLOUR CODING:

Black	Scar, Degeneration Suture Contact Lens	Red	Blood, Rose Bengal Superficial Vessel Stromal Vessel Ghost Vessel (dashes)
Blue	Edema Decemet's Fold (wavy lines) Epithelial Oedema (small circles)	Green	Fluorescein, Lens, Vitreous
Brown	Pigment, Iris	Yellow or Orange	Infiltrate, Hypopyon, Keratic Precipitates